nature photonics

Article

https://doi.org/10.1038/s41566-023-01171-7

Parallelized computational 3D video microscopy of freely moving organisms at multiple gigapixels per second

Received: 14 October 2022

Accepted: 3 February 2023

Published online: 20 March 2023

Check for updates

Kevin C. Zhou[1,2,6], Mark Harfouche[2], Colin L. Cooke[3], Jaehee Park[2], Pavan C. Konda[1], Lucas Kreiss[1], Kanghyun Kim[1], Joakim Jönsson[1], Thomas Doman[2], Paul Reamey[2], Veton Saliu[2], Clare B. Cook[1,2], Maxwell Zheng[2], John P. Bechtel[2], Aurélien Bègue[2], Matthew McCarroll[4], Jennifer Bagwell[5], Gregor Horstmeyer[2], Michel Bagnat[5] & Roarke Horstmeyer[1,2,3]

Wide-field-of-view microscopy that can resolve three-dimensional (3D) information at high speed and spatial resolution is particularly desirable for studying the behaviour of freely moving organisms. However, it is challenging to design an optical instrument that optimizes all these properties simultaneously. Existing techniques typically require the acquisition of sequential image snapshots to observe large areas or measure 3D information, thus compromising speed and throughput. Here we present 3D-RAPID, a computational microscope based on a synchronized array of 54 cameras that can capture high-speed 3D topographic videos over a 135 cm^2 area, achieving up to 230 frames per second at a spatiotemporal throughput exceeding 5 gigapixels per second. 3D-RAPID employs a 3D reconstruction algorithm that, for each synchronized snapshot, fuses all 54 images into a composite that includes a co-registered 3D height map. The self-supervised 3D reconstruction algorithm trains a neural network to map raw photometric images to 3D topography using stereo overlap redundancy and ray-propagation physics as the only supervision mechanism. The reconstruction process is thus robust to generalization errors and scales to arbitrarily long videos from arbitrarily sized camera arrays. We demonstrate the broad applicability of 3D-RAPID with several collections of freely behaving organisms: ants, fruit flies and zebrafish larvae.

Quantifying the behaviour and locomotion of freely moving organisms, such as the fruit fly and zebrafish, is essential in many applications, including neuroscience[1–3], developmental biology[4], disease modelling[5,6], drug discovery[7,8] and toxicology[9–13]. In particular for high-throughput screening, it is desirable to monitor tens or hundreds of organisms simultaneously, thus requiring high-speed imaging over large fields of view (FOVs) at high spatial resolution, ideally in three dimensions. Such an imaging system would enable researchers

[1]Department of Biomedical Engineering, Duke University, Durham, NC, USA. [2]Ramona Optics Inc., Durham, NC, USA. [3]Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. [4]Department of Pharmaceutical Chemistry, University of California, San Francisco, CA, USA. [5]Department of Cell Biology, Duke University, Durham, NC, USA. [6]Present address: Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, CA, USA. e-mail: kevinczhou@berkeley.edu; roarke.w.horstmeyer@duke.edu

https://doi.org/10.1038/s41566-023-01171-7

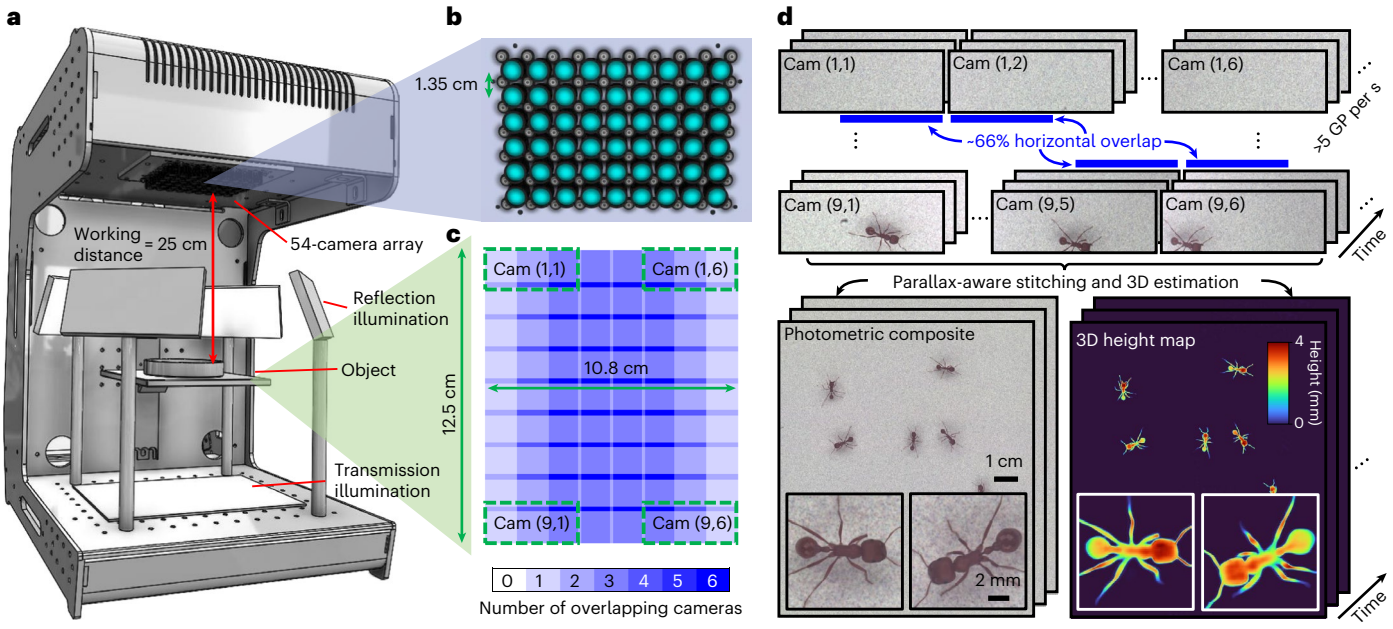


Fig. 1 | Overview of 3D-RAPID. a, Computational microscope setup, consisting of a 9 × 6 = 54 array of finite-conjugate imaging systems, jointly recording across an area of 135 cm^2. LED arrays serve as the illumination source, both in transmission and reflection. **b**, The 9 × 6 array of cameras and lenses. **c**, Overlap map of the object plane, demonstrating roughly 66% horizontal overlap redundancy between neighbouring cameras (and minimal overlap in the vertical dimension). Four example camera FOVs are denoted by the green dashed boxes, identified by their (row,column) coordinates. **d**, The MCAM captures 54 synchronized videos at throughputs of >5 GP per s, and these are stitched to form a high-speed video sequence of globally consistent composites and the corresponding 3D height maps.

to bridge the gap between microscopic phenotypic expression and multi-organism behaviour across more macroscopic scales, such as shoaling[14,15], courtship and aggression behaviours[16,17], exploration[18,19] and hunting[19–23].

Common approaches for behavioural recording use two-dimensional (2D) wide-field microscopes at low magnification to cover as large a FOV as possible. However, due to limitations in the space–bandwidth product (SBP) of conventional optics[24,25], in standard imaging systems there is a tradeoff between the image resolution and the FOV. Techniques that enhance the SBP to facilitate high-resolution imaging over large areas, such as Fourier ptychography[26] and mechanical sample translation[27,28], often require multiple sequential measurements, which compromises the imaging speed and throughput. Approaches that perform closed-loop mechanical tracking to record single organisms moving freely in two dimensions, using scanning mirrors[29] or moving cameras[19], do not scale to multiple organisms.

Conventional wide-field techniques also lack 3D information, which potentially precludes the observation of important behaviours, such as vertical displacement and out-of-plane tilt changes in zebrafish larvae[23,30,31] and 3D limb coordination and kinematics in various insects[32–35]. Commonly used 3D microscopy techniques, such as diffraction tomography[36,37], light sheet microscopy[38,39] and optical coherence tomography[40–42], are not well-suited for behavioural imaging since they often require multiple sequential measurements for 3D estimation and inertia-limited scanners that sacrifice speed. Furthermore, while such techniques can achieve micrometre-scale resolutions, they typically have millimetre-scale FOVs rather than the multi-centimetre-scale FOVs needed for imaging freely moving organisms. Thus, these techniques are typically limited to imaging one immobilized organism at a time (for example, embedded in agarose, tethered[34,35] or paralysed).

Camera-array-based imaging systems have also been proposed to increase the SBP and overall throughput[43–47]; however, none of these earlier approaches has demonstrated scalable, high-speed, high-resolution, wide-FOV, 3D imaging. In particular, several of these approaches were designed for 2D macroscopic photographic applications, which face challenges of miniaturization for microscopy applications, or feature a primary objective lens that limits the maximum achievable SBP. Macroscale 3D imaging techniques such as time-of-flight LiDAR (light detection and ranging)[48], coherent LiDAR[49–51], structured light[52], stereo vision[53] and active stereo vision techniques[54] have throughputs typically limited to tens of megapixels per second (MP per s) and have millimetre-scale spatial resolution.

Here we present 3D-RAPID (that is, 3D reconstruction with an array-based parallelized imaging device), a new computational 3D microscope based on an array of 9 × 6 = 54 synchronized cameras; this device is capable of acquiring continuous high-speed video of dynamic 3D topographies over a lateral FOV of 135 cm^2, with a 3D spatial resolution of tens of micrometres and a spatiotemporal data rate exceeding 5 gigapixels per second (GP per s) (Fig. 1). We demonstrate three operating modes, which can be flexibly chosen depending on whether the speed (up to 230 frames per second (f.p.s.)) or the spatial SBP (up to 146 MP per frame) is to be prioritized. We also present a new scalable 3D reconstruction algorithm that, for each synchronized snapshot, simultaneously forms a globally consistent photometric composite and a co-registered 3D height map according to a ray-based physical model. The 3D reconstruction itself trains a spatiotemporally compressed convolutional neural network (CNN) that maps multi-ocular inputs to the 3D topographies, using ray-propagation physics and consistency in the overlapped regions as the only supervision. Thus, after computational reconstruction of just a few video frames (<20), 3D-RAPID can rapidly generate photometric composites and 3D height maps for the remaining video frames non-iteratively.

3D-RAPID thus solves a long-standing problem in the field of behavioural imaging of freely moving organisms that previously only enabled low-throughput solutions. We demonstrated the broad applicability of 3D-RAPID in three model organisms: zebrafish, fruit flies and ants. In particular, the large FOV of 3D-RAPID enables the imaging of multiple freely behaving organisms in parallel, while the dynamic 3D reconstructions and high spatial resolution and imaging speeds enable

https://doi.org/10.1038/s41566-023-01171-7

Table 1 | The three imaging configurations

Downsample factor	1× (none)	2×	4×
Per-camera dimensions	1,536×4,096	768×2,048	384×1,024
Composite dimensions	13,000×11,250	6,500×5,625	3,250×2,810
Composite SBP (MP)	146.3	36.6	9.1
Frame rate (f.p.s.)	15	60	230
Exposure (ms)	20	5	2.5
Raw pixel rate (GP per s)	5.1	5.1	4.9
Composite pixel rate (GP per s)	2.2	2.2	2.1
Image pixel pitch (μm)	9.6	19.2	38.4

the 3D tracking of fine features, such as the joints of ant legs during exploration, the eye orientation of zebrafish larvae during feeding and the pose of fruit flies while grooming.

High-throughput 3D video with 3D-RAPID

3D-RAPID hardware design

The 3D-RAPID hardware is based on a multi-camera array microscope (MCAM) architecture[47,55], which consists of 54 synchronized micro-camera units spaced at 1.35 cm and tiled in a 9 × 6 configuration. Each micro-camera captures up to 3,120 × 4,208 pixels (1.1 μm pitch) for a total of ~700 MP per snapshot. The data are transmitted to the computer memory via PCIe (Peripheral Component Interconnect express) at ~5 GB s^{-1}. We axially positioned the lenses (f = 26.23 mm, Supply Chain Optics) to obtain a magnification of $M \approx 0.11$, leading to around a 66% overlap in the sample-plane FOV between adjacent cameras along the longer camera dimension (Fig. 1c), so that almost every point in the synthesized ~12.5 × 10.8 cm area is viewed at least twice. This overlap redundancy enables 3D estimation using stereoscopic parallax cues. The sample is illuminated in transmission or reflection using planar arrays of white light-emitting diodes (LEDs) covered by diffusers (Fig. 1a).

Tradeoff space of lateral resolution, FOV and frame rate

3D-RAPID has the flexibility to downsample or crop sensors or use fewer cameras to increase the frame rate. The overall data throughput is limited by the slower of two factors: the data-transfer rate from the sensors to the computer memory (~5 GB s^{-1}) or the sensor readout rate, both of which are functions of the sensor crop shape and downsample factor. Streaming all 54 cameras without downsampling or cropping runs into the data-transfer rate-limited frame rate of ~7 f.p.s. To achieve higher frame rates, we present results with a 1,536 × 4,096 sensor crop using either 4×, 2× or no downsampling, enabling frame rates of 230, 60 or 15 f.p.s., respectively, while maintaining roughly the same throughput of ~5 GP per s (Table 1). While excluding half of the sensor rows nearly eliminates FOV overlap in the vertical dimension, the benefits are twofold: increased frame rate and reduced rolling-shutter artefacts (see 'Temporal synchronization of the camera array' in the Methods).

Seamless image registration, stitching and 3D estimation

For each video frame, the 3D-RAPID algorithm fuses the 54 synchronously acquired images, via gradient descent using a pixel-intensity-based loss, into a continuous, seamless, expanded-FOV composite image, and simultaneously estimates a co-registered 3D height map (Fig. 2a). In fact, these two tasks are intimately related: to form a high-quality registration, it is necessary to account for parallax distortions induced by height deviations from a planar sample scene that would otherwise thwart simple registration using homographic transformations (Fig. 2b)[56,57]. To achieve this, the algorithm starts with calibration of the six-degree-of-freedom poses (x, y, z, roll, pitch, yaw), camera distortions and intensity variations by registering and stitching 54 images of a flat, patterned target (see 'Camera calibration: pose, distortion and intensity variation' in the Methods). Estimating the 3D height map of the sample of interest relative to this calibration plane is tantamount to rendering the images registerable using homographies (Fig. 2b). In particular, the per-pixel deformation vectors that undo the parallax shifts (that is, orthorectify the images) have magnitudes that are directly proportional to the per-pixel height values, $h(\mathbf{r})$ (that is, the height map), given by[57]

$$h(\mathbf{r}_{\text{obj}} + \mathbf{r}_{\text{rectify}}) = f\frac{\| \mathbf{r}_{\text{rectify}} \|}{\| \mathbf{r}_{\text{obj}} - \mathbf{r}_{\text{vanish}} \|}\left(1 + \frac{1}{M}\right) \quad (1)$$

where f = 26.23 mm is the effective focal length of the lens, $M \approx 0.11$ is the linear magnification, $\mathbf{r}_{\text{obj}}$ is the apparent 2D position of the object in the pixel (before orthorectification), $\mathbf{r}_{\text{vanish}}$ is the vanishing point to which all lines perpendicular to the sample plane appear to converge and $\mathbf{r}_{\text{rectify}}$ is the 2D orthorectification vector pointing towards the vanishing point (Fig. 2b). The parameter $\mathbf{r}_{\text{vanish}}$ can be determined from the camera pose, as the point in the sample plane that intersects with the perpendicular line that passes through the principal point in the thin-lens model. The orthorectification vectors $\mathbf{r}_{\text{rectify}}$, and therefore the height map, for each object position $\mathbf{r}_{\text{obj}}$ can be determined by registering images (via photometric pixel values) from different perspectives. The accuracy of the height map thus depends on the object having photometrically textured (that is, not uniform) surfaces that enable unique image registration, a condition satisfied by all model organisms we imaged.

Thus, the optimization problem is to jointly register all 54 images using the pixel-wise photometric loss, using the orthorectification maps (which are directly proportional to the height maps via equation (1)) as the deformation model on top of the fixed, pre-calibrated camera parameters, including distortions (Fig. 2a,b). In practice, since the viewpoint-dependent photometric appearance can affect the image registration, we also employed normalized high-pass filtering to standardize the photometric appearance (see 'Achieving robustness to illumination variation' in the Methods and Supplementary Section 3.5).

Spatiotemporally compressed 3D video via physics-supervised learning

Instead of optimizing the height maps directly, we re-parameterized them as the output of an encoder–decoder CNN that takes the multi-view stereo images as inputs. This re-parameterization has two interpretations, depending on whether we emphasize the CNN or the ray-based physical model. First, the CNN can be thought to act entirely as a training-data-free regularizer (that is, a deep image prior (DIP)[58]) that safeguards against 3D reconstruction artefacts that may otherwise arise from practical deviations from modelling assumptions that thwart image registration[57]. For example, using the CNN as a regularizer can be useful when the sample has a different appearance when viewed from different angles, which can be caused by uneven illumination, angle-dependent scattering responses or varying pixel responses. Since we wish to reconstruct hundreds to thousands of 3D video frames, it would be prohibitively slow to independently reconstruct every video frame, with or without CNNs. Thus, we use one shared DIP, with each frame encoded by the raw multi-ocular stereo photometric inputs.

Thus, the second interpretation is one of physics-supervised learning, in which the image registration of the overlapped MCAM image frames, governed by a ray-based thin-lens physical model (equation (1)), provides the physics-based supervision that guides the CNN training (Fig. 2a,c). The CNN can then be used to generalize to other MCAM data, both spatially (other micro-cameras) and temporally (other video frames).

This dual interpretation of our CNN-regularized, physics-supervised learning approach reveals several advantages. First, since

https://doi.org/10.1038/s41566-023-01171-7

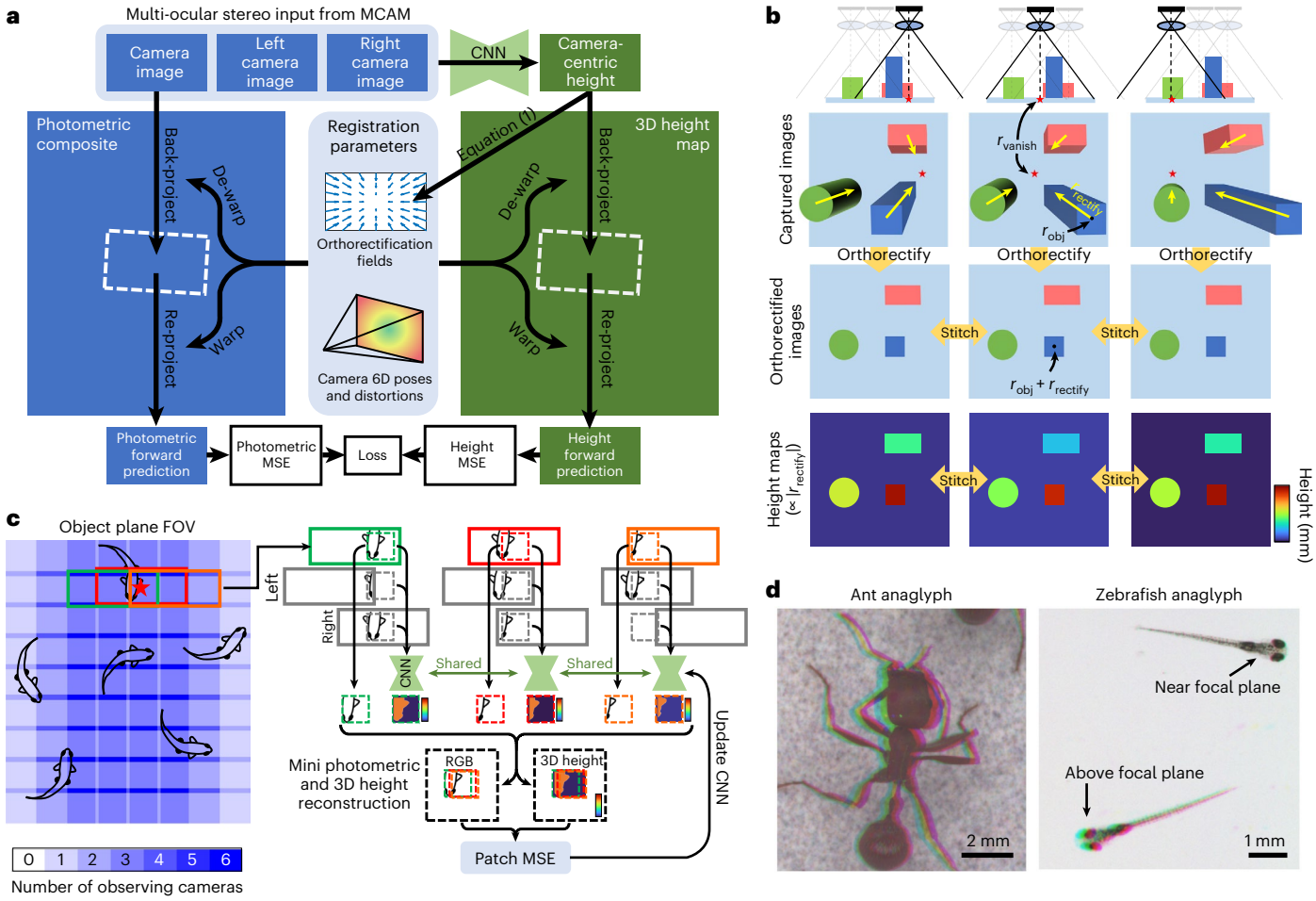


Fig. 2 | Computational 3D reconstruction and stitching algorithm for 3D-RAPID. a, The algorithm starts with raw red, green and blue (RGB) images (only one shown for clarity), along with co-registered images from the cameras to left and right, as CNN inputs. The CNN generates camera-centric height maps, which in turn dictate orthorectification fields (see **b** and equation (1)). Orthorectificaton fields and camera poses plus distortions constitute registration parameters, dictating where and how each image should be back-projected in the stitched photometric composite and 3D height map. The back-projection step is then reversed (re-projection) to form forward predictions of the RGB images and camera-centric height maps. Errors (the photometric mean squared error (MSE) and the height MSE) guide the optimization of the CNN. **b**, The physical ray model, intuitively showing how orthorectification facilitates the stitching of non-telecentric images and height maps. **c**, The patch-based joint training/stitching/3D reconstruction algorithm. At each gradient-descent iteration, random coordinates are chosen (red star); all cameras that view a given point are isolated. A patch is cropped out from each camera image surrounding the randomly sampled point, along with the corresponding left/right camera images to serve as the multi-ocular stereo inputs to the CNN to predict the patch height map. These patches undergo the procedure outlined in **a** to form a mini photometric and 3D height reconstruction to update the CNN. Values of zero are assigned to stereo input pixels when unavailable (for example, at the edge of the object-plane FOV), to preserve convolutionality when applying the CNN to the entire camera image to generate the full-size reconstructions. **d**, Anaglyphs, in which the three stereo inputs are colour coded as RGB channels, showing the parallax that is used to estimate three dimensions.

we use a fully convolutional CNN, we can optimize arbitrarily sized image patches (Fig. 2c) that can fit in the GPU (graphics processing unit) memory and then perform non-iterative forward inference on arbitrarily large full-size images (Supplementary Fig. 4). Thus, our proposed approach is scalable and generalizable to arbitrarily many cameras, each with arbitrarily many pixels, for arbitrarily many video frames. For implementation details on patch-based training, see 'Patch-based learning from multi-ocular stereo inputs', Fig. 2c and Supplementary Section 3. Second, the CNN enforces a spatiotemporally compressed representation of the 3D height-map videos, thus avoiding the need to iteratively optimize each frame individually. Third, this spatiotemporal compression reduces the chances of overfitting, as there are far fewer CNN parameters than height-map pixels across all video frames. Furthermore, the CNN implicitly enforces consistency across space and time, thus, for example, avoiding variance induced by independent optimization runs on different frames. Fourth, our approach has an inherent fail-safe against generalization errors, unlike other deep-learning-based approaches, since the ground truth is always available via the overlap redundancy and the physical model.

Patch-based learning from multi-ocular stereo inputs

Although Fig. 2a summarizes the ideal joint 3D reconstruction, stitching and training method, in practice we are constrained by the GPU memory. Thus, we train the CNN using a random patch-sampling approach (Fig. 2c). Briefly, at each optimization iteration, we sample n_{batch} (batch size) random points within the composite FOV (one is shown in Fig. 2c). All cameras viewing each point are selected, from which patches surrounding that point are extracted from each camera view. Thereafter, these n_{batch} groups of selected patches independently undergo the procedure outlined in Fig. 2a. Once CNN training is done, the back-projection step in Fig. 2a is carried out for each full temporal frame to create the stitched RGBH 3D reconstructions (containing red,

https://doi.org/10.1038/s41566-023-01171-7

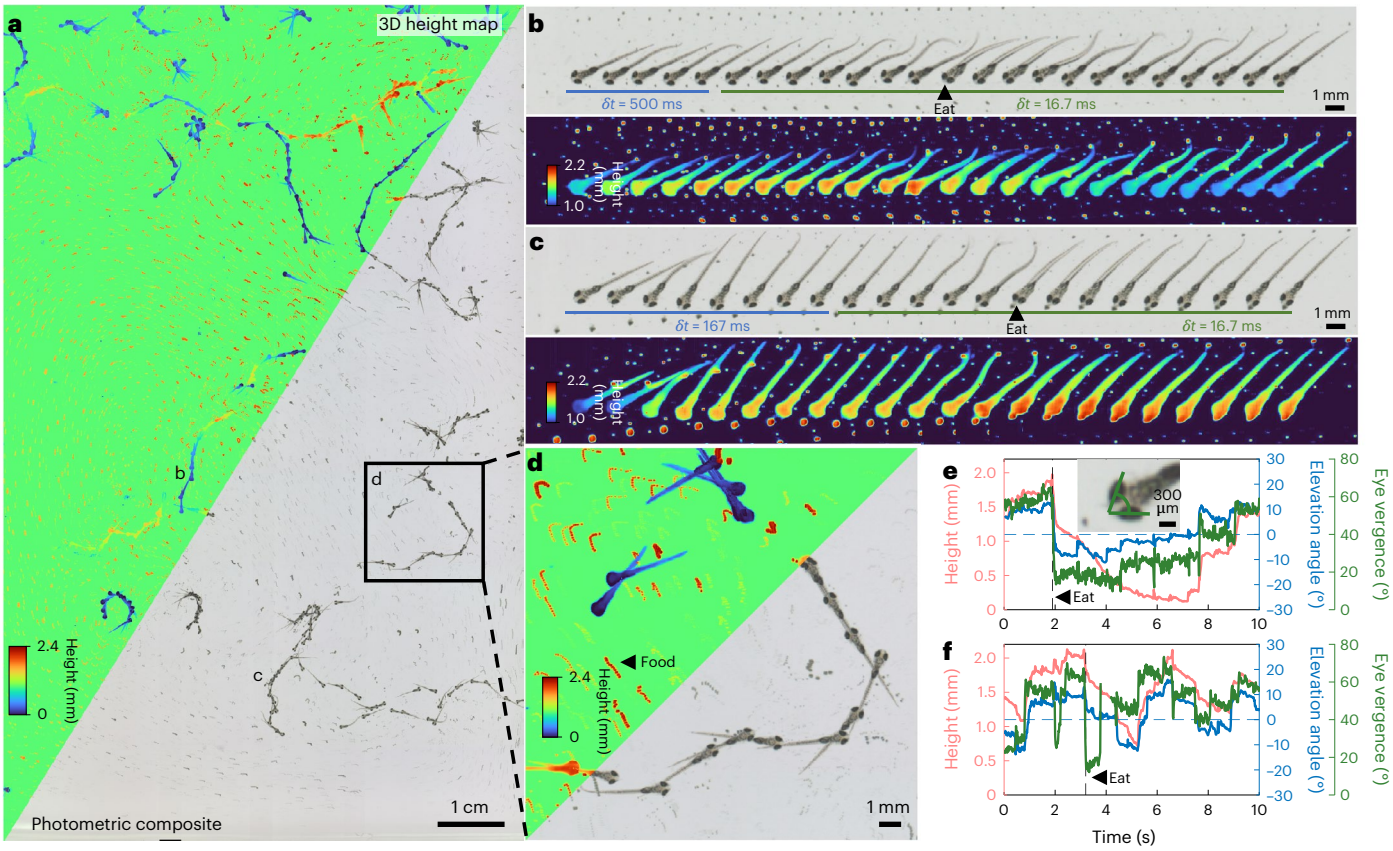


Fig. 3 | Zebrafish larvae (10 dpf) swimming in an open arena with interspersed microcapsule food particles (AP100), acquired at 60 f.p.s. for 10 s (Supplementary Videos 1 and 3). a, 3D height map and photometric composite of the zoomed-out FOV, projected across every 50th temporal frame (0.83 s) to highlight the dynamics. The height map assigns an arbitrary value to the otherwise empty background. **b**,**c**, Photometric (top) and height map (bottom) frames of single tracked fish feeding on AP100 particles in which the first five (**b**) and eight (**c**) frames are spaced at $\delta t = 500$ ms (**b**) and 167 ms (**c**) intervals and the remaining frames are spaced at 16.7 ms intervals (the full frame rate). **e**,**f**, Head height, elevation angle (pitch) and eye-vergence angle (as illustrated in the inset of **e**) of the same fish in **b** and **c**, respectively, throughout the 10 s video. Note the change in eye vergence before and after the feedifng event in both **b** and **c**. **d**, A zoomed-in region of **a**, showing three individual larvae in varying states of activity. The small red tracks are the drifting and floating AP100 food particles. Population-level analysis is summarized in Extended Data Fig. 1.

green, blue and height channels) (Supplementary Fig. 4). For more implementation details, see Supplementary Section 3.

Results

3D-RAPID system characterization

Our 3D-RAPID system has a full-pitch lateral resolution of ~25 µm and a depth of field of ~9.4 mm, based on imaging a USAF resolution target and translating a patterned target axially (see Supplementary Section 1). We validated the height precision and accuracy of our 3D-RAPID system through imaging precisely machined (to within 0.3 µm) gauge blocks (Mitutoyo). As expected, the accuracy and precision of the reconstructed height are improved at higher spatial resolution, which facilitates the more accurate measurement of parallax shifts (Supplementary Section 5). Specifically, we achieved sub-20 µm accuracy and precision in the 15 f.p.s. configuration, and ~37 µm accuracy and ~74 µm precision in the 230 f.p.s. configuration. See Supplementary Section 1 for detailed characterization.

Zebrafish (*Danio rerio*) larvae

We applied 3D-RAPID to several 10 s videos of zebrafish larvae swimming freely in a large 97 × 130 mm open arena using the 60 f.p.s. and 230 f.p.s. configurations (Table 1) across three separate experiments, the first of which was on fish at 10 days post-fertilization (dpf) feeding on microcapsule food particles (AP100) (Supplementary Videos 1 (60 f.p.s.), 2 (230 f.p.s.) and 3 (60 f.p.s. with tracking)). Figure 3 and Extended Data Fig. 1 summarize the results for the 60 f.p.s. video of the 10 dpf fish feeding on AP100, most of which are floating at or near the water surface (Extended Data Fig. 1b). We tracked all 40 fish using a simple particle-tracking algorithm (see 'Organism tracking and pose determination' in the Methods; Supplementary Video 3). The high throughput of 3D-RAPID enables us to observe fine detail over a very wide FOV (Fig. 3a–d), capturing multiple rapid-feeding events (around tens of milliseconds), as shown in Fig. 3b,c. From the photometric images, we can see that the larvae turn their bodies laterally so that their ventrally positioned mouths can access the overhead floating food. We also observe their eye convergence once the larvae identify and approach the target[20–22]. The eye angles rapidly deconverge after food capture (Fig. 3e,f). The older fish (20 dpf) exhibit similar eye behaviour when feeding on brine shrimp (Supplementary Videos 4 and 5).

The 3D information accessed using our technique reveals how the larvae axially approach their targets from below, including their head heights and elevation (pitch) angles during these feeding events (Fig. 3b,c and e,f, respectively)[23]. The larvae's head height matches that of the targeted food particle during ingestion (see also Supplementary Videos 1, 2, 4 and 5), offering a validation of our technique.

In addition to making organism-level observations, the high throughput of 3D-RAPID enables us to make population-level inferences by aggregating the height and elevation-angle information for all 40 individually tracked larvae for all in-frame time points. The results show a roughly linear trend between height and elevation angle

https://doi.org/10.1038/s41566-023-01171-7

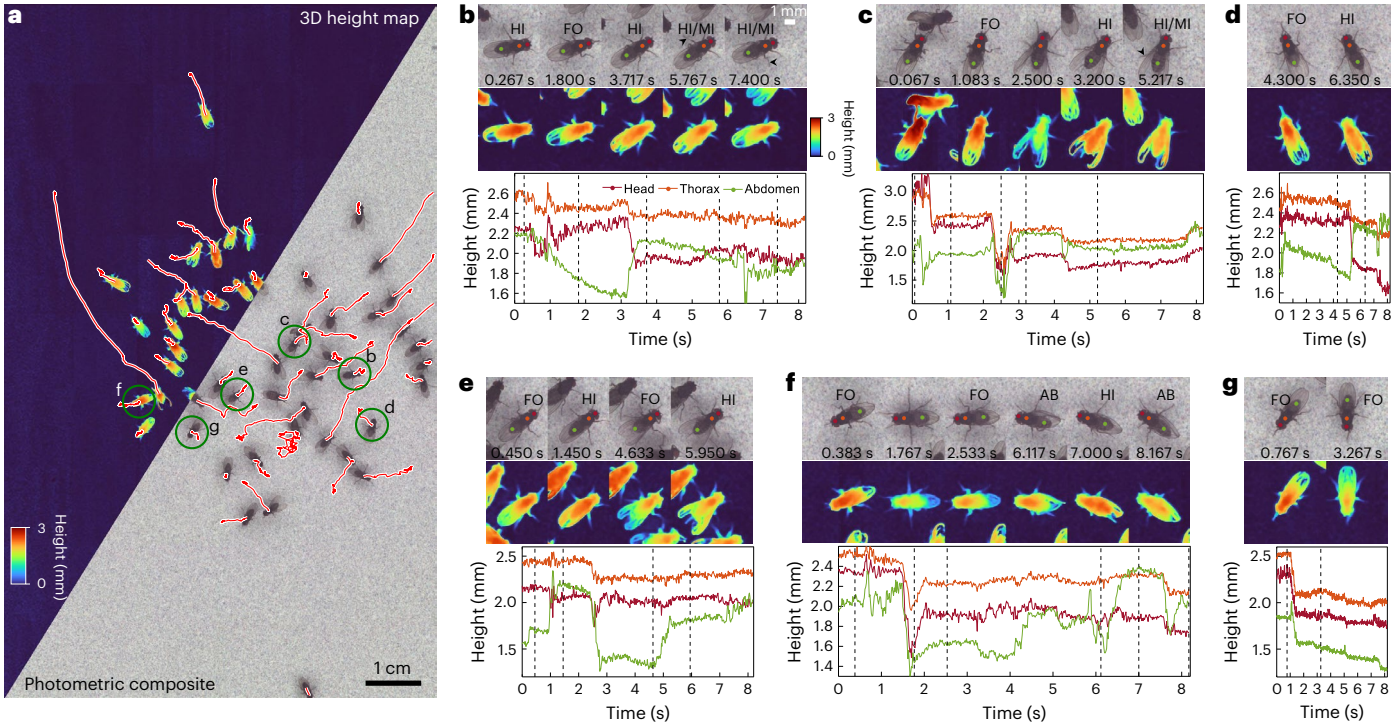


Fig. 4 | Adult fruit flies freely moving across a flat, noise-patterned surface, acquired at 60 f.p.s. for 8 s (Supplementary Videos 8 and 10). a, 3D height map and photometric composites of the zoomed-out FOV. The white-outlined red lines are the trajectories taken by the 50 flies. The green-circled flies are analysed in the other figure panels. **b**, Selected photometric (top) and height-map (bottom) frames of a single tracked fly, as indicated in **a**, exhibiting several grooming behaviours. HI, hindleg grooming; FO, foreleg/head grooming; MI, middle leg participation; AB, abdominal grooming. The time points of the frames are indicated by dashed lines in the plot below, which in turn highlights the changing heights of the head, thorax and abdomen for the different grooming actions. **c**–**g**, Selected photometric (top) and height map (bottom) frames of five additional tracked flies (as indicated in **a**) exhibiting several grooming behaviours, where the colour scale of the 3D height maps is the same as that in **b**. Population-level analysis is summarized in Extended Data Fig. 2.

(Extended Data Fig. 1a), which can be explained based on the mobility constraints defined by the length of the larvae and the water depth. For example, if the head is at the bottom of the arena, then the elevation angle must be negative. Assuming a larval length of $L = 4$ mm and a water depth of $H = 2.3$ mm, these geometric constraints on the elevation angle, ϕ, for a fish at height, h, are

$$\phi_{\min}(h) = \sin^{-1}(h/L), \quad \phi_{\max}(h) = \sin^{-1}((H-h)/L), \qquad (2)$$

which are plotted in Extended Data Fig. 1a. This offers additional validation of the accuracy of our 3D height maps. We also estimated the probability distributions of the heights of the larvae and the food particles (Extended Data Fig. 1b), both of which are bimodal. Predominantly, the larvae dwell at the bottom of the arena, only occasionally venturing upwards for food.

In addition, we analysed population-level correlations between the eye-vergence angle (see 'Organism tracking and pose determination' in the Methods)—a property observable in the photometric images—and the fish height and elevation angle, which are derived from our 3D height maps (Extended Data Fig. 1c,d), across $n = 39$ fish (one stationary fish was excluded). Specifically, we used a linear mixed-effects model, where the height or elevation angle is the fixed effect and any dependence among images from the same fish is accounted for via random effects. Analyses of variance suggest that while the fish height is not a statistically significant linear predictor of the eye-vergence angle ($p = 0.33$) the fish elevation angle is ($p < 10^{-5}$). This is consistent with the fact that when the fish is swimming upwards, it is probably focusing on a food particle close to the surface. By contrast, the fish can still be close to the surface following a feeding event, immediately after which the eyes deconverge (Fig. 3b,c,e,f).

With the 230 f.p.s. configuration, we can trade off spatial resolution to temporally resolve higher-speed zebrafish larval locomotion. For example, compare the beginning of Supplementary Videos 6 and 7, which feature rapidly swimming zebrafish larvae, captured at 60 f.p.s. and 230 f.p.s., respectively. Similarly, we can resolve the 3D fish dynamics as it attempts to swallow a live brine shrimp (Supplementary Videos 4 (60 f.p.s.) and 5 (230 f.p.s.)).

Fruit flies (*Drosophila hydei*)

Next, we applied 3D-RAPID to image 50 freely exploring adult fruit flies under the 60 f.p.s. (Supplementary Videos 8 and 10) and 230 f.p.s. (Supplementary Video 9) configurations. Figure 4 and Extended Data Fig. 2 summarize the results for the 60 f.p.s. configuration for six individual flies exhibiting various grooming behaviours. Supplementary Video 10 shows the tracking of all 50 flies. The 3D height map reveals changes in the fly height and body tilt as the flies transition between different grooming behaviours. In Fig. 4b, as the fly transitions between grooming using its hindlegs and forelegs, the abdomen moves up and down, respectively. When a middle leg joins the grooming (Fig. 4b, black arrowheads in the photometric frames), there is a subtle change in the abdomen height relative to the head height. In Fig. 4c, our method correctly predicts an elevated height as one fly climbs on top of another. At 2.500 s, the fly's height drops, consistent with the straightened leg joints. A similar body tilt trend is observed for foreleg versus hindleg grooming in this fly, as is also shown in Fig. 4d–f. In Fig. 4f, we see another instance of the fly's leg joints being fully extended at 1.767 s, resulting in a reduced overall height. Furthermore, we observe that the abdomen takes on a different relative height during abdominal grooming compared with hindleg grooming. Finally, in Fig. 4g, although the fly is grooming using its forelegs throughout

https://doi.org/10.1038/s41566-023-01171-7

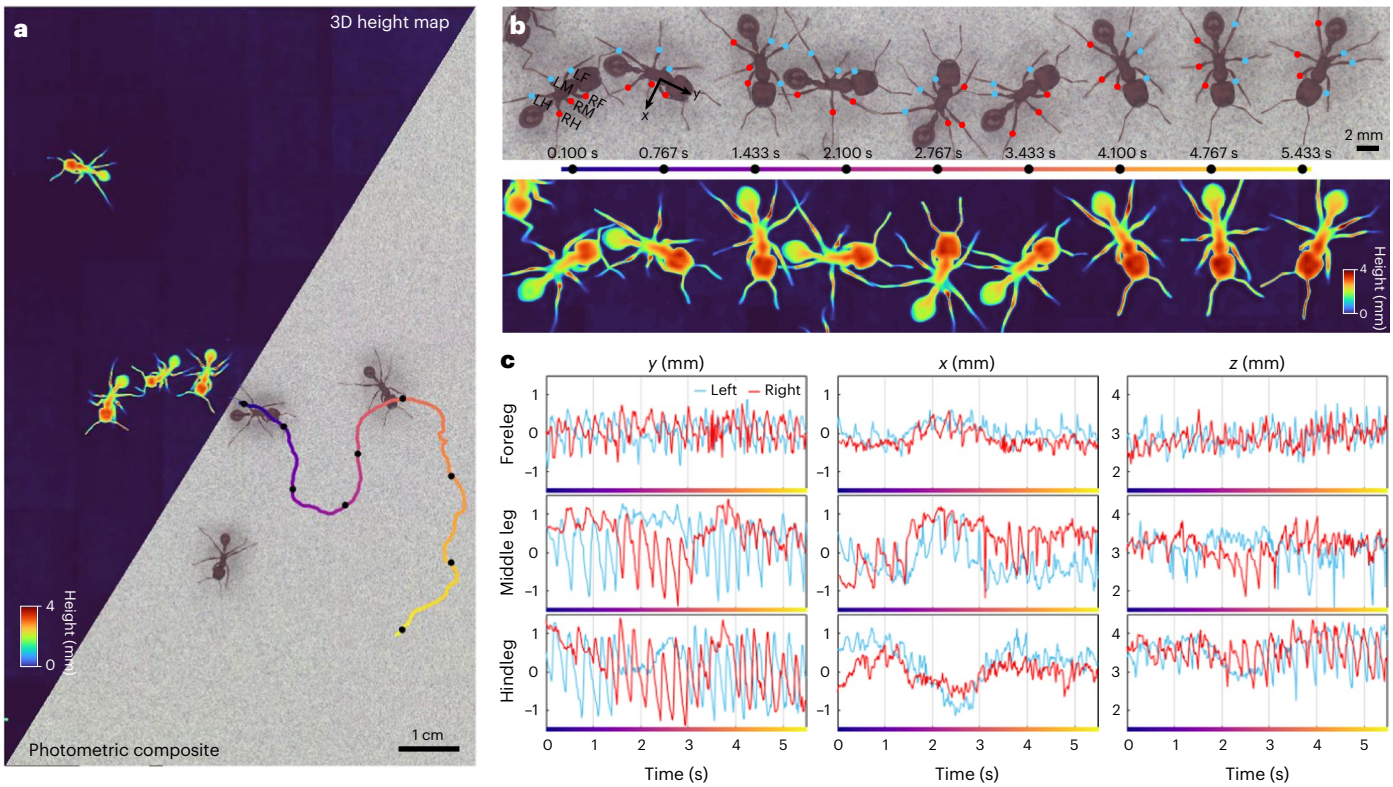


Fig. 5 | Harvester ants freely moving across a flat, noise-patterned surface, acquired at 60 f.p.s. for 10 s (Supplementary Videos 11 and 12). **a**, Photometric composite and 3D height map of the zoomed-out FOV. The trajectory of one of the ants is colour coded by time, progressing from purple to yellow over a duration of 5.5 s, and is analysed in **b** and **c**. **b**, Photometric (top) and height-map (bottom) frames showing temporal snapshots of a single tracked ant along the trajectory in **a**. The blue and red dots indicate the femur–tibia joints of the ant's six legs. L, left; R, right; F, foreleg; M, middle leg; H, hindleg. **c**, The 3D (x, y, and z) positions of the femur–tibia joints over the 5.5 s trajectory. The lateral dimensions (x–y) are defined relative to the ant's orientation, as illustrated in **b**.

the video, it reduces its overall height after about 1 s, consistent with its extended leg posture.

To analyse population trends, we annotated video frames across $n = 43$ flies flies with one of five behaviours: hindleg grooming, foreleg/head grooming, abdomen grooming, standing still, and walking (Extended Data Fig. 2). Flies that exited the FOV were excluded. We tested for cross-behavioural differences in heights of the head, thorax and abdomen using three separate linear categorical mixed-effects models, accounting for random effects due to correlations among video frames from the same fly. Analyses of variance suggest that the behaviour groups are a statistically significant predictor of the heights of the head ($p < 10^{-7}$), thorax ($p < 10^{-16}$) and abdomen ($p < 10^{-62}$).

Harvester ants (*Pogonomyrmex barbatus*)

We also imaged freely exploring red harvester ants under the 60 f.p.s. (Supplementary Video 11) and 230 f.p.s. (Supplementary Video 12) configurations. The 60 f.p.s. results are summarized in Fig. 5. We used the dynamic 3D reconstructions to track the femur–tibia joints of all six legs of an individual ant (Fig. 5b,c; section 'Organism tracking and pose determination' in the Methods), providing information about the kinematics of ant locomotion. The joint trajectories are plotted in Fig. 5c, showing that the high-frequency (~3–4 Hz) oscillations from walking kinematics are anti-correlated between left and right legs. This frequency remains relatively constant throughout the ant's journey. Furthermore, the forelegs and hindlegs on the same side of the body are correlated, but anti-correlated with the middle legs on the same side of the body. These behaviours are consistent with the well-known alternating tripod gait pattern in ants[33,59,60], which persists even as the curvature of the ant trajectory changes.

We also observe changes in lower-frequency gait patterns as the ant makes multiple turns throughout its exploration. In the first ~1.5 s, as the ant is turning right, we see a reduced oscillation amplitude in the middle and hindlegs on the right side in both the y and z directions; however, for the x direction we see the opposite trend (see Fig. 5b for the ant-centric coordinate system). Between 1.5 and 3 s, as the ant is turning left, we see the opposite motions as in the first 1.5 s—the oscillation amplitudes in the middle and hindlegs on the left are reduced in both the y and z directions, while the amplitude of the right middle leg motion in the x direction is reduced. From 3 to 4.5 s, the ant once again is turning right and we see similar trends as in the first 1.5 s. Overall, this reduction in motion in the y and z directions on the side of the ant corresponding to the direction the ant is turning is consistent with earlier knowledge[59]. Interestingly, the amplitudes of the foreleg oscillations on both the left and right sides in both the y and z directions remain relatively constant throughout the entire 5.5 s, suggesting a lesser role in the biomechanics of changing direction.

Finally, we observe a low-frequency oscillation (with a period of ~4 s) in the x direction for all six legs that is correlated with the curvature of the ant's trajectory. Unlike the high-frequency (3–4 Hz) walking kinematics, which are anti-correlated between left and right, these low frequencies are correlated between left and right legs, suggesting left–right coordination when the ant is turning. Furthermore, these low frequencies in the x direction are correlated between the forelegs and middle legs but are anti-correlated with the hindlegs.

Discussion

We have presented 3D-RAPID, a new computational microscope with a unique capability of dynamic topographic 3D imaging at a resolution of tens of micrometres, over a >130 cm^2 FOV and at a throughput

exceeding 5 GP per s. To handle this high data throughput, we devised an efficient, end-to-end, physics-supervised, CNN-based, joint 3D reconstruction-and-stitching algorithm that scales to arbitrarily long videos and arbitrarily sized camera arrays. Thus, 3D-RAPID fills a unique niche, enabling scientists to study several unconstrained, freely behaving model organisms in parallel at high speed and in high resolution over a very large FOV.

3D-RAPID differs from other camera-array-based techniques[43–47] in several ways, due to the dense camera-packing requirements necessitated by the high magnifications associated with microscopy. Some approaches alleviate this challenge by first magnifying to an intermediate image plane. However, the intrinsic SBP of the primary objective will limit the total throughput. Instead, our approach tiles complementary metal–oxide–semiconductor (CMOS) sensors on a common printed circuit board (PCB), which is connected to a single field-programmable gate array (FPGA) for unified data routing. This enables extremely tight packing and scalability by simply adding more sensors.

Although we have presented several convincing 3D behavioural imaging demonstrations, there are several avenues for improvement. The hardware parameters could be adjusted to improve the 3D height reconstruction accuracy, which depends on how accurately parallax shifts can be detected to match features from different cameras (see Supplementary Section 5). Furthermore, since the reconstruction algorithm is agnostic to the contrast mechanism, it would also be possible to incorporate, for example, fluorescence to correlate behaviours with molecular signatures. Finally, the throughput could be improved beyond 5 GP per s by alleviating data-transfer bottlenecks.

In summary, we have presented 3D-RAPID as a new platform for studying the behaviour of multiple freely moving organisms at high speed and resolution over a very wide area. We expect our technique to be broadly applicable for elucidating new behavioural phenomena, not only in zebrafish, fruit flies and ants but also in other model organisms such as tadpoles and nematodes.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41566-023-01171-7.

References

1. Bellen, H. J., Tong, C. & Tsuda, H. 100 years of *Drosophila* research and its impact on vertebrate neuroscience: a history lesson for the future. *Nat. Rev. Neurosci.* **11**, 514–522 (2010).
2. Oliveira, R. F. Mind the fish: zebrafish as a model in cognitive social neuroscience. *Front. Neural Circuits* **7**, 131 (2013).
3. Kalueff, A. V., Stewart, A. M. & Gerlai, R. Zebrafish as an emerging model for studying complex brain disorders. *Trends Pharmacol. Sci.* **35**, 63–75 (2014).
4. Dreosti, E., Lopes, G., Kampff, A. R. & Wilson, S. W. Development of social behavior in young zebrafish. *Front. Neural Circuits* **9**, 39 (2015).
5. Pandey, U. B. & Nichols, C. D. Human disease models in *Drosophila melanogaster* and the role of the fly in therapeutic drug discovery. *Pharmacol. Rev.* **63**, 411–436 (2011).
6. Sakai, C., Ijaz, S. & Hoffman, E. J. Zebrafish models of neurodevelopmental disorders: past, present, and future. *Front. Mol. Neurosci.* **11**, 294 (2018).
7. MacRae, C. A. & Peterson, R. T. Zebrafish as tools for drug discovery. *Nat. Rev. Drug Discov.* **14**, 721–731 (2015).
8. Maitra, U. & Ciesla, L. Using *Drosophila* as a platform for drug discovery from natural products in Parkinson's disease. *Medchemcomm* **10**, 867–879 (2019).
9. Hirsch, H. V. et al. Behavioral effects of chronic exposure to low levels of lead in *Drosophila melanogaster*. *Neurotoxicology* **24**, 435–442 (2003).
10. Bambino, K. & Chu, J. Zebrafish in toxicology and environmental health. *Curr. Top. Dev. Biol.* **124**, 331–367 (2017).
11. Rihel, J. et al. Zebrafish behavioral profiling links drugs to biological targets and rest/wake regulation. *Science* **327**, 348–351 (2010).
12. McCarroll, M. N. et al. Zebrafish behavioural profiling identifies GABA and serotonin receptor ligands related to sedation and paradoxical excitation. *Nat. Commun.* **10**, 4078 (2019).
13. Mathias, J. R., Saxena, M. T. & Mumm, J. S. Advances in zebrafish chemical screening technologies. *Future Med. Chem.* **4**, 1811–1822 (2012).
14. Wright, D. & Krause, J. Repeated measures of shoaling tendency in zebrafish (*Danio rerio*) and other small teleost fishes. *Nat. Protoc.* **1**, 1828–1831 (2006).
15. Harpaz, R., Nguyen, M. N., Bahl, A. & Engert, F. Precise visuomotor transformations underlying collective behavior in larval zebrafish. *Nat. Commun.* **12**, 6578 (2021).
16. Dankert, H., Wang, L., Hoopfer, E. D., Anderson, D. J. & Perona, P. Automated monitoring and analysis of social behavior in *Drosophila*. *Nat. Methods* **6**, 297–303 (2009).
17. Robie, A. A., Seagraves, K. M., Egnor, S. R. & Branson, K. Machine vision methods for analyzing social interactions. *J. Exp. Biol.* **220**, 25–34 (2017).
18. Dunn, T. W. et al. Brain-wide mapping of neural activity controlling zebrafish exploratory locomotion. *Elife* **5**, 12741 (2016).
19. Johnson, R. E. et al. Probabilistic models of larval zebrafish behavior reveal structure on many scales. *Curr. Biol.* **30**, 70–82 (2020).
20. Bianco, I. H., Kampff, A. R. & Engert, F. Prey capture behavior evoked by simple visual stimuli in larval zebrafish. *Front. Syst. Neurosci.* **5**, 101 (2011).
21. Patterson, B. W., Abraham, A. O., MacIver, M. A. & McLean, D. L. Visually guided gradation of prey capture movements in larval zebrafish. *J. Exp. Biol.* **216**, 3071–3083 (2013).
22. Muto, A. & Kawakami, K. Prey capture in zebrafish larvae serves as a model to study cognitive functions. *Front. Neural Circuits* **7**, 110 (2013).
23. Bolton, A. D. et al. Elements of a stochastic 3D prediction engine in larval zebrafish prey capture. *Elife* **8**, 51975 (2019).
24. Lohmann, A. W. Scaling laws for lens systems. *Appl. Opt.* **28**, 4996–4998 (1989).
25. Park, J., Brady, D. J., Zheng, G., Tian, L. & Gao, L. Review of bio-optical imaging systems with a high space–bandwidth product. *Adv. Photonics* **3**, 044001 (2021).
26. Zheng, G., Horstmeyer, R. & Yang, C. Wide-field, high-resolution Fourier ptychographic microscopy. *Nat. Photonics* **7**, 739–745 (2013).
27. Kumar, N., Gupta, R. & Gupta, S. Whole slide imaging (WSI) in pathology: current perspectives and future directions. *J. Digit. Imaging* **33**, 1034–1040 (2020).
28. Borowsky, A. D. et al. Digital whole slide imaging compared with light microscopy for primary diagnosis in surgical pathology a multicenter, double-blinded, randomized study of 2045 cases. *Arch. Pathol. Lab. Med.* **144**, 1245–1253 (2020).
29. Grover, D., Katsuki, T. & Greenspan, R. J. Flyception: imaging brain activity in freely walking fruit flies. *Nat. Methods* **13**, 569–572 (2016).
30. Ehrlich, D. E. & Schoppik, D. Control of movement initiation underlies the development of balance. *Curr. Biol.* **27**, 334–344 (2017).
31. Ehrlich, D. E. & Schoppik, D. A primal role for the vestibular sense in the development of coordinated locomotion. *Elife* **8**, 45839 (2019).

32. Akitake, B. et al. Coordination and fine motor control depend on *Drosophila* TRPγ. *Nat. Commun.* **6**, 7288 (2015).
33. Shamble, P. S., Hoy, R. R., Cohen, I. & Beatus, T. Walking like an ant: a quantitative and experimental approach to understanding locomotor mimicry in the jumping spider *Myrmarachne formicaria*. *Proc. R. Soc. B* **284**, 20170308 (2017).
34. Günel, S. et al. DeepFly3D, a deep learning-based approach for 3D limb and appendage tracking in tethered, adult *Drosophila*. *Elife* **8**, 48571 (2019).
35. Lobato-Rios, V. et al. NeuroMechFly, a neuromechanical model of adult *Drosophila melanogaster*. *Nat. Methods* **19**, 620–627 (2022).
36. Wolf, E. Three-dimensional structure determination of semi-transparent objects from holographic data. *Opt. Commun.* **1**, 153–156 (1969).
37. Chowdhury, S. et al. High-resolution 3D refractive index microscopy of multiple-scattering samples from intensity images. *Optica* **6**, 1211–1219 (2019).
38. Chen, B.-C. et al. Lattice light-sheet microscopy: imaging molecules to embryos at high spatiotemporal resolution. *Science* **346**, 1257998 (2014).
39. Patel, K. B. et al. High-speed light-sheet microscopy for the in-situ acquisition of volumetric histological images of living tissue. *Nat. Biomed. Eng.* https://doi.org/10.1038/s41551-022-00849-7 (2022).
40. Huang, D. et al. Optical coherence tomography. *Science* **254**, 1178–1181 (1991).
41. Zhou, K. C., Qian, R., Dhalla, A.-H., Farsiu, S. & Izatt, J. A. Unified k-space theory of optical coherence tomography. *Adv. Opt. Photonics* **13**, 462–514 (2021).
42. Zhou, K. C. et al. Computational 3D microscopy with optical coherence refraction tomography. *Optica* **9**, 593–601 (2022).
43. Wilburn, B. et al. High performance imaging using large camera arrays. *ACM Trans. Graph.* **24**, 765–776 (2005).
44. Brady, D. J. et al. Multiscale gigapixel photography. *Nature* **486**, 386–389 (2012).
45. Lin, X., Wu, J., Zheng, G. & Dai, Q. Camera array based light field microscopy. *Biomed. Opt. Express* **6**, 3179–3189 (2015).
46. Fan, J. et al. Video-rate imaging of biological dynamics at centimetre scale and micrometre resolution. *Nat. Photonics* **13**, 809–816 (2019).
47. Thomson, E. E. et al. Gigapixel imaging with a novel multi-camera array microscope. *Elife* **11**, e74988 (2022).
48. Jiang, Y., Karpf, S. & Jalali, B. Time-stretch LiDAR as a spectrally scanned time-of-flight ranging camera. *Nat. Photonics* **14**, 14–18 (2020).
49. Riemensberger, J. et al. Massively parallel coherent laser ranging using a soliton microcomb. *Nature* **581**, 164–170 (2020).
50. Rogers, C. et al. A universal 3D imaging sensor on a silicon photonics platform. *Nature* **590**, 256–261 (2021).
51. Qian, R. et al. Video-rate high-precision time-frequency multiplexed 3D coherent ranging. *Nat. Commun.* **13**, 1476 (2022).
52. Geng, J. Structured-light 3D surface imaging: a tutorial. *Adv. Opt. Photonics* **3**, 128–160 (2011).
53. Aguilar, J.-J., Torres, F. & Lope, M. Stereo vision for 3D measurement: accuracy analysis, calibration and industrial applications. *Measurement* **18**, 193–200 (1996).
54. Scharstein, D. & Szeliski, R. High-accuracy stereo depth maps using structured light. In *Proc. 2003 IEEE Computer Society Conference on Computer Vision and Pattern Recognition* 195–202 (IEEE, 2003).
55. Harfouche, M. et al. Imaging across multiple spatial scales with the multi-camera array microscope. *Optica* **10**, (2023); https://doi.org/10.1364/OPTICA.478010
56. Kumar, R., Anandan, P. & Hanna, K. Direct recovery of shape from multiple views: a parallax based approach. In *Proc. 12th International Conference on Pattern Recognition* 685–688 (IEEE, 1994).
57. Zhou, K. C. et al. Mesoscopic photogrammetry with an unstabilized phone camera. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 7535–7545 (IEEE, 2021).
58. Ulyanov, D., Vedaldi, A. & Lempitsky, V. Deep image prior. In *Proc. IEEE Conference on Computer Vision and Pattern Recognition* 9446–9454 (IEEE, 2018).
59. Zollikofer, C. Stepping patterns in ants – influence of speed and curvature. *J. Exp. Biol.* **192**, 95–106 (1994).
60. Reinhardt, L. & Blickhan, R. Level locomotion in wood ants: evidence for grounded running. *J. Exp. Biol.* **217**, 2358–2370 (2014).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2023

Methods

Temporal synchronization of the camera array

Ideally, all sensor pixels should be fully synchronized with a global shutter, not only within each sensor but also across sensors. This will ensure that, between different views of the same object, after accounting for camera poses, the only discrepancies are due to parallax shifts and not sample motion. For example, if two camera views of a moving object with zero height were desynchronized, lateral motion could be interpreted as a parallax shift, leading to an erroneous height estimate. In practice, each of our sensors exhibits a rolling shutter, whereby only a single pixel value can be read out at a given time for a given sensor, row by row from the top-left to the bottom-right corner in a raster scan pattern. This means that the bottom of a given sensor is captured later than the top of the sensor immediately below. However, across independent sensors, this rolling-shutter readout pattern is synchronized to within 10 µs, limited by the serial communication interface (I2C with a 100 kHz clock).

To mitigate the rolling-shutter effects, we employed two strategies. First, we cropped the sensors so that there is only substantial overlap in the horizontal dimension for stitching, in which the desynchronization is much less severe. Second, we calculated that, with exposures of 2.5 ms for 4× downsampling, 5 ms for 2× downsampling and 20 ms for no downsampling, artefacts would be minimal. For a detailed discussion and calculations, see Supplementary Section 4.

Achieving robustness to illumination variation

Since the optimization metric of our approach is the mean squared per-pixel photometric error, we will achieve an optimal performance when the sample has a camera-independent photometric appearance. This condition will require not only a uniform response across all pixels of all cameras but also that the sample is isotropically emanating light in all directions. The latter property is in practice difficult to achieve, requiring either perfectly diffuse illumination or a diffusely scattering sample, regardless of the illumination direction. In addition to the regularizing effects of the CNN/DIP, we used two further strategies to reduce the effects of a camera-dependent appearance. First, as part of the camera pose pre-calibration procedure, we also jointly optimized per-camera second-order 2D polynomials (with cross terms) to correct the slowly varying image intensity variation (whether caused by uneven illumination or camera response), using the same photometric stitching loss. Thus, the pre-calibration step ensures not only the geometric consistency of the 54 images but also photometric continuity. For more details, see 'Camera calibration: pose, distortion and intensity variation' below.

Second, for terrestrial organisms illuminated in reflection, we used a two-step optimization process, where we first optimize the CNN to register the images using the RGB intensities. In the second step, we continue optimizing the CNN, except this time registering normalized high-pass-filtered versions of the photometric images, which reduces illumination-induced differences in the photometric appearance and emphasizes edges (Supplementary Section 3.5). This two-step procedure effectively removes artefacts in the 3D height maps that would otherwise result from camera-dependent photometric appearances.

Camera calibration: pose, distortion and intensity variation

The first step in the 3D estimation pipeline was to calibrate the geometric and photometric properties of the cameras. Specifically, the geometric properties include their six-degree-of-freedom pose (the 3D position plus the 3D orientation) and second-order radial distortions (for example, pincushion or barrel distortions). The photometric properties include the pixel-intensity variations both within individual cameras and across different cameras. These may arise due to vignetting, uneven illumination, pixel response variation or angle-dependent scattering of the sample. To estimate the calibration parameters, we imaged a flat, epi-illuminated, homogeneously patterned calibration target using the MCAM and registered the resulting 54 images, enforcing both geometric and photometric consistency in the overlapped regions.

The calibration procedure follows the optimization procedure outlined in Fig. 2a, excluding the height-map-related orthorectification portion. In particular, let $\mathbf{x}_0$ and $\mathbf{y}_0$ be two vectors representing the ideal 2D spatial coordinates of the camera pixels—that is, a 2D rectangular grid of equally spaced points (for example, 1,536 × 4,096). Next, let $D_\theta\{\cdot,\cdot\}$ be an image deformation operation that maps from the ideal camera coordinates to a common global coordinate space (the object plane), parameterized by the camera parameters θ. See Supplementary Section 1 of ref. [57] for specific implementation details of D_θ. Let θ_i be the camera parameters for the ith camera, so that

$$\mathbf{x}_i, \mathbf{y}_i = D_{\theta_i}\{\mathbf{x}_0, \mathbf{y}_0\} \tag{3}$$

represents the (de)warped coordinates of the ith camera in a common object plane.

Let $\mathbf{I}_{i,0}$ be a vector of the same length as x_0 and y_0, indicating the measured photometric intensity at every pixel coordinate for the ith camera. Although the debayered images have three colour channels, here, for simplicity, we assume a single-channel image. Furthermore, let $C_{\phi,\mathbf{x}_0,\mathbf{y}_0}\{\cdot\}$ be a photometric correction operation, parameterized by ϕ, so that

$$\mathbf{I}_i = C_{\phi_i,\mathbf{x}_0,\mathbf{y}_0}\{\mathbf{I}_{i,0}\} \tag{4}$$

represents the photometrically adjusted intensity values for the ith camera. The dependence on $\mathbf{x}_0$ and $\mathbf{y}_0$ indicates that the photometric correction is spatially varying. Specifically, we used a second-order polynomial correction,

$$\begin{aligned}\mathbf{I}_i = C_{\phi_i,\mathbf{x}_0,\mathbf{y}_0}\{\mathbf{I}_{i,0}\} = \big(&a_{i,0} + a_{i,1}\mathbf{x}_0 + a_{i,2}\mathbf{y}_0 + a_{i,3}\mathbf{x}_0 \odot \mathbf{x}_0 + \\ &a_{i,4}\mathbf{y}_0 \odot \mathbf{y}_0 + a_{i,5}\mathbf{x}_0 \odot \mathbf{y}_0\big) \odot \mathbf{I}_{i,0},\end{aligned} \tag{5}$$

where $\odot$ represents element-wise multiplication and $\phi_i = \{a_{i,0}, a_{i,1}, a_{i,2}, a_{i,3}, a_{i,4}, a_{i,5}\}$. In sum, assuming that θ_i and ϕ_i are optimized, then $\{\mathbf{x}_i, \mathbf{y}_i, \mathbf{I}_{i,0}\}$ represents the corrected ith camera data, accounting for distortion and photometric variation.

Next, let $\{\mathbf{x}, \mathbf{y}, \mathbf{I}\}$ be three vectors representing the flattened concatenation of $\{\mathbf{x}_i, \mathbf{y}_i, \mathbf{I}_{i,0}\}$ for all i. We then initialize a blank matrix, $\mathbf{R}[\cdot,\cdot]$, representing the stitched reconstruction, into which we back-project the collection of points,

$$\mathbf{R}[\mathbf{x}, \mathbf{y}] \leftarrow \mathbf{I}, \tag{6}$$

with interpolation, as $\mathbf{x}$ and $\mathbf{y}$ are continuously valued. When specific coordinates are visited more than once, the values are averaged. The result of equation (6) is an estimate of the stitched composite for a given set of $\{\theta_i, \phi_i\}_{i=1}^{54}$. To update these parameters, we form a forward prediction from $\mathbf{R}[\cdot,\cdot]$ by re-projecting back into the camera spaces, as follows:

$$\mathbf{I}_{\text{pred}} = \mathbf{R}[\mathbf{x}, \mathbf{y}]. \tag{7}$$

$\mathbf{I}_{\text{pred}}$ should match $\mathbf{I}$ when the camera images are well registered and the corrected photometric intensities match in overlapped regions. Thus, we minimize the error metric,

$$\text{MSE} = \| \mathbf{I}_{\text{pred}} - \mathbf{I} \|^2, \tag{8}$$

with respect to $\{\theta_i, \phi_i\}_{i=1}^{54}$ via gradient descent. Since the image target is homogeneous, we also include a regularization term,

$$\sum_i \sigma(\mathbf{I}_i), \tag{9}$$

which enforces a homogeneous reconstruction. Here, the standard deviation operation ($\sigma(\cdot)$) is taken across all the pixels in one image.

Finally, we apply the calibrated parameters, $\{\theta_i, \phi_i\}_{i=1}^{54}$, to each frame of the videos of the freely moving organisms. To homogenize the background in the case of the zebrafish, which uses transmission illumination instead of the epi-illuminated calibration target, we apply a second calibration step that optimizes only the photometric correction parameters, $\{\phi_i\}_{i=1}^{54}$, using the maximum projection of the video across time, which eliminates all moving objects.

Organism tracking and pose determination

To track the fruit flies, zebrafish larvae and harvester ants, we first thresholded the photometric composites to segment each organism and compute each of their centroids across all video frames. We then employed a simple particle-tracking algorithm, matching the organisms by finding the closest centroid in the subsequent video frame. In the case of clashing match proposals, we assigned matches that minimized the sum of the total absolute lateral displacements. To track the six femur–tibia joints of the ants we incorporated the observation that the joint heights are local maxima in the 3D height maps for segmentation, and employed a similar particle-tracking algorithm.

To determine the orientation of the organisms, we performed principal component analysis on the thresholded pixel coordinates and took the first principal component (PC) as the organism's orientation. In the case of the zebrafish, we used the height-map coordinates to perform principal component analysis in three dimensions, thereby enabling us to compute the elevation angles in Fig. 3. We resolved the sign ambiguity of the PC either by enforcing the dot products of the PCs of the tracked organism in consecutive frames to be positive or by computing the relative displacement between the unweighted centroid and the intensity-weighted centroid and forcing the PC to point in the same direction.

The fish eye-vergence angles were estimated by thresholding the green channel of the photometric intensity images to identify the eyes. The orientations of the eyes were estimated using the regionprops command in MATLAB, which finds the angle of the major axis of the ellipse with the equivalent second moments. The vergence angle is then computed as the angle between the two eyes.

Biological samples and data acquisition

Zebrafish stocks were bred and maintained following the guidelines of the Duke University Institutional Animal Care Use Committee and as previously described[61]. Zebrafish were stored at 28 °C with daily feeding and water changes, and cycled through 14 h of light and 10 h of darkness each day. Free swimming fish were imaged at larval stages between 5 dpf and 20 dpf. Specifically, zebrafish larvae were transferred from culture chambers using a transfer pipette to a clear plastic imaging arena (with lateral inner dimensions of 97 × 130 mm), which was filled with system water a few millimetres deep. The arena was then placed on the sample stage of the MCAM system. The *z* position of the stage was adjusted such that the zebrafish larvae were all within the depth of field of the lenses. The system was left undisturbed with the LED illumination panels turned on for at least 5 min to allow the zebrafish to acclimatize, after which multiple MCAM videos were acquired using a custom Python script. After video acquisition, the arena was removed and replaced with a flat patterned calibration target. We focused the target with the *z* stage using a Laplacian-based sharpness metric and captured a single frame (all 54 cameras), which served to calibrate the camera poses and distortions for all videos captured during that imaging session.

The wild-type red harvester ants and fruit flies (available from various vendors on Amazon) were maintained at room temperature. When ready for imaging, we positioned and focused a flat patterned calibration target, which serves two purposes: (1) for camera calibration, just as for the zebrafish videos described in the previous paragraph and (2) to serve as a flat substrate for the ants and fruit flies to walk upon. The patterned target, although not required, serves as a global reference in the 3D height maps. Alternatively, the substrate could be monochrome/featureless or transparent (for example, a glass sheet), as was the case for the zebrafish imaging configuration, in which case the 3D height map would assign an arbitrary height value to the background without affecting the 3D accuracy of the organisms themselves.

The ants or fruit flies were inserted into a Falcon tube and released onto the centre of the flat substrate, after which we immediately ran the same custom Python script to acquire MCAM videos. If necessary, the insects were re-collected in the tubes and re-released into the arena for repeated imaging. After video acquisition, we acquired a single frame of the calibration target alone, just as we did after zebrafish video acquisition.

Data availability

The data that support the findings of this study are available from the Duke Research Data Repository at https://doi.org/10.7924/r4db86b1q. Interactive, full-resolution, reconstructed video frames can be viewed at https://gigazoom.rc.duke.edu/.

Code availability

The Python code used to generate the high-resolution 3D videos featured in this study is available at https://github.com/kevinczhou/3D-RAPID.

References

61. Westerfield, M. *The Zebrafish Book: a Guide for the Laboratory Use of Zebrafish (*Danio rerio*)* 4th edn (Univ. Oregon Press, 2000); https://zfin.org/zf_info/zfbook/zfbk.html

Acknowledgements

We would like to thank K. Branson, S. Turaga, T. Dunn, A. Chakraborty and M. Hoffmann for their helpful feedback on the manuscript. Research reported in this publication was supported by the Office of Research Infrastructure Programs (ORIP), Office of the Director, National Institutes of Health of the National Institutes of Health and the National Institute of Environmental Health Sciences (NIEHS) of the National Institutes of Health under award number R44OD024879 (M.H., J.P., T.D., P.R., V.S., M.Z., J.P.B., A.B. and G.H.), the National Cancer Institute (NCI) of the National Institutes of Health under award number R44CA250877 (M.H., J.P., T.D., P.R., V.S., M.Z., J.P.B., A.B. and G.H.), the National Institute of Biomedical Imaging and Bioengineering (NIBIB) of the National Institutes of Health under award number R43EB030979 (M.H., J.P., T.D., P.R., V.S., M.Z., J.P.B., A.B. and G.H.), the National Science Foundation under award number 2036439 (M.H., J.P., T.D., P.R., V.S., M.Z., J.P.B., A.B. and G.H.) and the Duke Coulter Translational Partnership Award (K.C.Z., K.K. and R.H.).

Author contributions

K.C.Z. and R.H. conceived the idea and initiated the research. With the help of C.L.C., J.P., P.C.K. and R.H., K.C.Z. developed the algorithms and theory. K.C.Z. wrote the code for and performed 3D video reconstruction and stitching, animal tracking and data analysis. M.H., T.D., P.R., V.S., C.B.C., M.Z. and R.H. developed the MCAM hardware and acquisition software. With the help of J.P.B., J.B., A.B., G.H., K.K. and R.H., K.C.Z. acquired and analysed the biological data. M.M., J.B. and M.B. provided input to and supervision of the biological experiments. T.D., J.J., K.K. and K.C.Z. created the supplementary videos and visualizations. K.C.Z. wrote the manuscript and created the figures, with input from all authors. K.C.Z, L.K. and R.H. revised the manuscript. R.H. supervised the research.

Competing interests

R.H. and M.H. are cofounders of Ramona Optics, Inc., which is commercializing multi-camera array microscopes. M.H., J.P., T.D., P.R.,

V.S., C.B.C., M.Z., J.P.B. and G.H. are or were employed by Ramona Optics, Inc. during the course of this research. K.C.Z. is a consultant for Ramona Optics, Inc. The remaining authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41566-023-01171-7.

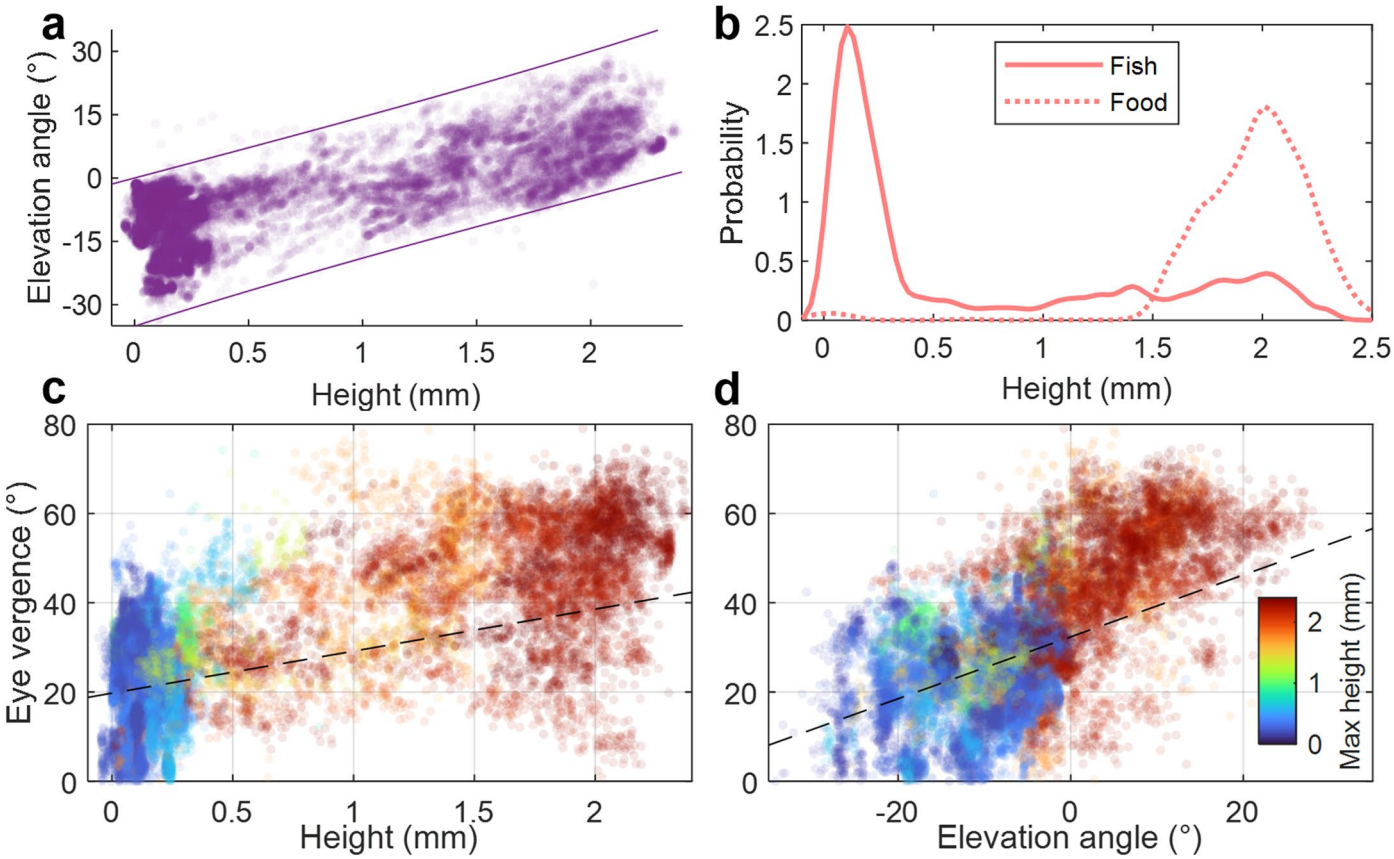
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41566-023-01171-7.

Correspondence and requests for materials should be addressed to Kevin C. Zhou or Roarke Horstmeyer.

Peer review information *Nature Photonics* thanks Chao Zuo and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

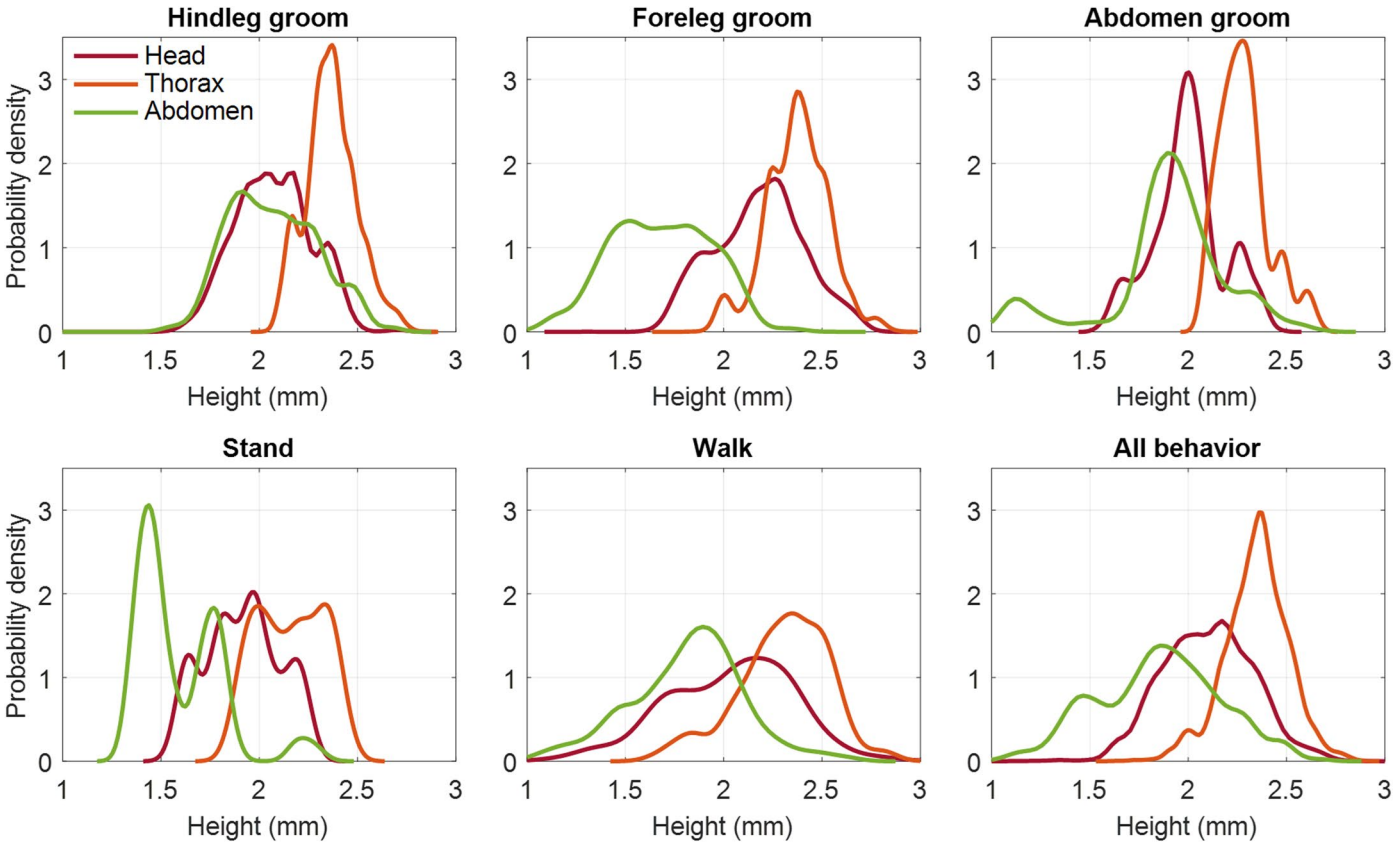
Reprints and permissions information is available at www.nature.com/reprints.

https://doi.org/10.1038/s41566-023-01171-7



Extended Data Fig. 1 | Population-level analysis of the zebrafish larvae featured in Fig. 3 and Supplementary Videos 1,3. a Fish head height vs. elevation angle for all 40 fish over time. Lines define the approximate physical limits due to geometric fish mobility constraints. **b** Kernel density estimates of the height distributions of the zebrafish and AP100 food particles. Eye vergence vs. head height (**c**) and vs. elevation angle (**d**) plots are color-coded by the maximum height the fish attained in the 10-sec video. Fixed effect components of the linear mixed-effects regression lines are plotted ($p = 0.33$ and $p < 10^{-5}$) for **c** and **d**, respectively.

https://doi.org/10.1038/s41566-023-01171-7



Extended Data Fig. 2 | Population-level analysis of the adult fruit flies featured in Fig. 4 and Supplementary Videos 8,10. The six plots show kernel densities of the heights of the head, thorax, and abdomen for various behaviors. Differences of head ($p < 10^{-7}$), thorax ($p < 10^{-16}$), and abdomen ($p < 10^{-62}$) heights across behaviors are statistically significant (n = 43 flies).